Water Footprint of India's E20 Transition: Impacts on Groundwater and Pathways for Sustainable Ethanol Production

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1. Introduction

India raised the ethanol share in petrol from about 1.5% in 2013-14 to about 19% in 2024-25, and brought the E20 target (20% ethanol by volume) forward from 2030 to 2025 (Fig. 1). Most of this ethanol comes from irrigated crops, so the transition moves water as well as energy. Published footprints are quoted as litres of water per litre of ethanol at national scale, which says little about whether the water comes from a recharged basin or an already over-drawn aquifer. **Research gap:** footprints, groundwater assessments and farm-gate prices are published separately and rarely read together. We ask: (i) how much hidden water does one litre of E20 petrol carry under each feedstock; (ii) how stressed are the aquifers where feedstocks are grown; and (iii) do price signals favour the crops with the largest footprint?

2. Methods

We compiled published datasets (metadata analysis) and combined them with simple, transparent calculations:

- **Footprints (L water per L ethanol):** NITI Aayog / MoPNG roadmap figures as reported by India Today and Outlook (2025): sugarcane 3,630, maize 4,670, rice 10,790; ICAR-IISR gives about 2,000 for sugarcane. Distillery process water is only 3-5 L per L (GEMA / MoPNG), so crop water dominates.
- **Hidden water in E20:** 0.2 L ethanol × footprint, assuming one feedstock supplies all ethanol.
- **Groundwater stress:** CGWB / Ministry of Jal Shakti Dynamic Ground Water Resource Assessment 2024; stage of extraction = annual extraction ÷ annual extractable resource.
- **Price signal and blending:** CACP / DA&FW MSP and sugarcane FRP, 2015-16 vs 2024-25 (change calculated by us); PIB / MoPNG / PPAC blending data (rounded); Agroprojectus (India) for drip irrigation.

3. Results

Aquifer context. India extracts 245.64 bcm of groundwater a year against 406.19 bcm extractable: a national stage of extraction of 245.64 ÷ 406.19 = 60.47%. Punjab (156.87%) and Haryana (135.96%), with Rajasthan and Delhi, exceed 100%, meaning they withdraw more than is recharged; Uttar Pradesh and Tamil Nadu are semi-critical (70-90%). Karnataka (66.3%) and Maharashtra (53.8%), major sugarcane states, were below 70% in the 2023 CGWB state table.

Footprint by feedstock (Fig. 2). Rice ethanol needs 10,790 L/L, 2.97 times sugarcane (3,630) and 2.31 times maize (4,670). Per litre of E20 petrol this is 726 L (sugarcane), 934 L (maize) and 2,158 L (rice), about 36, 47 and 108 twenty-litre buckets; with the ICAR-IISR estimate sugarcane falls to about 400 L.

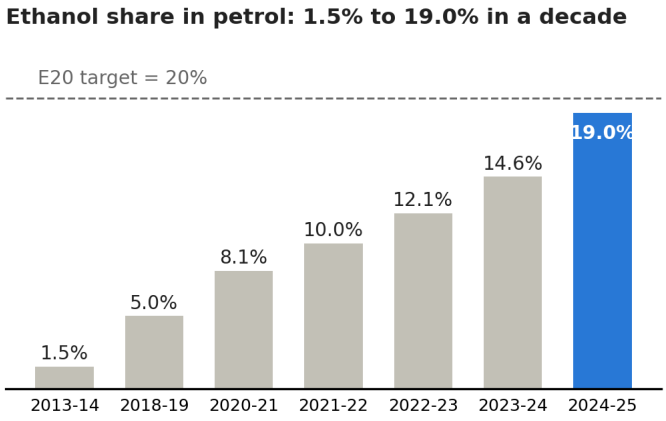


Fig. 1. PIB / MoPNG / PPAC blending data (rounded).

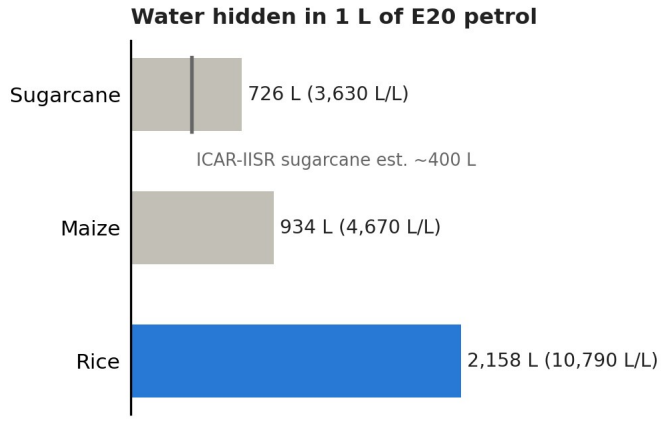


Fig. 2. Footprints: NITI Aayog / MoPNG via India Today and Outlook (2025); hidden water = 0.2 × footprint (calculated).

Price signal (Fig. 3). From 2015-16 to 2024-25, sugarcane FRP rose from ₹230 to ₹340 per quintal (+47.8%), maize MSP from ₹1,325 to ₹2,225 (+67.9%) and paddy MSP from ₹1,410 to ₹2,300 (+63.1%); the two crops with the largest footprints (rice, maize) saw the steepest rises. **Mitigation.** Drip irrigation cuts the sugarcane ethanol footprint from 2,469 to 1,634 L/L: (2,469 − 1,634) ÷ 2,469 = 33.8%.

Feedstock	Footprint (L water per L ethanol)	Hidden water per L of E20 petrol (L)	× sugarcane	Support-price rise, 2015-16 to 2024-25
Sugarcane	3,630	726	1.00	+47.8% (FRP)
Maize	4,670	934	1.29	+67.9% (MSP)
Rice (paddy price)	10,790	2,158	2.97	+63.1% (MSP)

Table 1. Summary of the calculated indicators (inputs as cited in Methods; all other columns calculated by us).

4. Discussion

A low litres-per-litre value does not by itself mean low risk. Sugarcane has the smallest footprint of the three crops, but it is grown across states with very different aquifer conditions, so the same litre of water has different consequences in Maharashtra than in over-exploited north-western states. Rice is the most water-intensive feedstock and is grown where aquifers are already over-drawn. Higher support prices for water-intensive crops can lock this in by rewarding what farmers already grow, whatever the aquifer says.

Five pathways follow: (1) micro-irrigation for cane; (2) water-budgeted distillery licensing that matches plant capacity to local recharge; (3) price incentives that favour lower-water feedstocks where aquifers are stressed; (4) crop residues and 2G ethanol, which add fuel without adding cropland; and (5) public groundwater and water-use disclosure for every plant. No single crop or technology fits every state; food security, farmer incomes and yields must be weighed too.

Recommendation. Feedstock and siting decisions for ethanol should be made against local aquifer status, not national averages: where extraction is above 100% of the extractable resource, even a low-footprint crop adds to an unsustainable draw, while in states below 70% the same crop poses far less risk. Our calculations give a simple screening rule: pair each state's stage of extraction with the footprint and price signal of the feedstock being promoted.

Limitations. Footprints vary by method and region and our values come from secondary reports. State averages hide district hotspots. We ran no statistical test between feedstock area and water-table decline, so the findings show co-location of risk, not cause; rainfall deficit, industry and urban demand also drive depletion. Next steps: district-level groundwater trends and controls for rainfall and industry.

5. Supplementary information: data and calculations

- **Inputs:** footprints (3,630 / 4,670 / 10,790 L/L; ICAR-IISR about 2,000 L/L for sugarcane); national groundwater extraction 245.64 bcm and extractable resource 406.19 bcm (CGWB 2024); Punjab 156.87% and Haryana 135.96%; MSP/FRP values in ₹ per quintal for 2015-16 and 2024-25.
- **Formulas:** hidden water = 0.2 × footprint; stage of extraction = extraction ÷ extractable resource × 100; price change = (P2024-25 ÷ P2015-16 − 1) × 100; drip saving = (conventional − drip) ÷ conventional.
- **Reproducibility:** every derived value can be reproduced in a spreadsheet from the inputs above; no values were estimated or assumed beyond those cited.

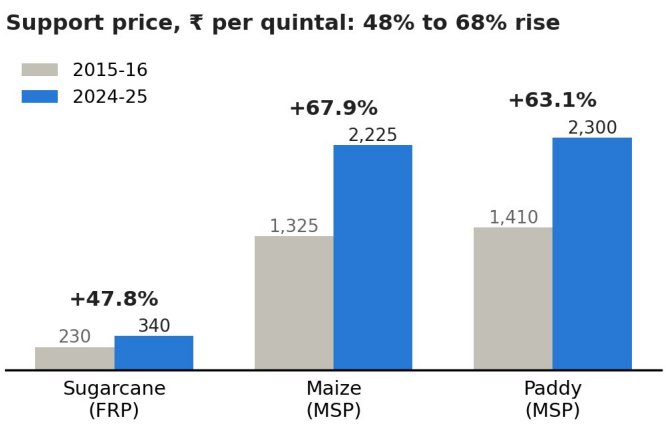


Fig. 3. CACP / DA&FW MSP notifications and sugarcane FRP orders; % change calculated by us.

Sources

NITI Aayog; MoPNG; ICAR-IISR; GEMA; CGWB / Ministry of Jal Shakti, Dynamic Ground Water Resources 2024 (national and Punjab reports) and 2023 state-wise table; CACP; DA&FW; PIB / PPAC; Agroprojectus (India); India Today and Outlook (2025). Footprint figures are secondary-source values; blending percentages are rounded. All derived values (hidden water, price changes, stage of extraction, drip saving) are our own calculations from the cited inputs.